

Javier Chang (常中嶺)

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Skills

- **Firmware:** C, Arduino, UART / Serial Protocols
- **Backend:** .NET (C# / .NET Core), PHP (Laravel), Python (Django), C
- **Frontend & Mobile:** Vue.js (RWD), Flutter, Android (Java), UWP
- **Database & DevOps:** MSSQL, MySQL, Enterprise Virtualization (vSphere / Horizon VDI)

Patent

- M455937 - Intelligent LED-Guided Directional Moving Floor Escape System
- M487500 - Smart Escape Route Guidance and Directional Indicator System

Work Experience

Institute for Information Industry (III) | Full-Stack Engineer

2023.12 - Present

- Led system analysis (SA), technical documentation, and MSSQL schema design for the "New Performance Management System"; systematically resolved cross-departmental edge cases, reducing routine communication and maintenance overhead between IT and HR by 40%.
- Implemented a decoupled MVC architecture using .NET Core and Vue.js; designed high-concurrency Stored Procedures for large-scale data matching, replication, and core CRUD business logic, establishing rigorous validation to ensure data consistency.
- Developed an end-to-end online approval workflow module (including dynamic nodes for draft, submission, rejection, withdrawal, and approval), fully digitizing traditional paper-based processes to eliminate delays and archival inefficiencies.
- Managed legacy system maintenance, code refactoring, and comprehensive QA testing; successfully diagnosed logical flaws in legacy performance and manager-evaluation subsystems, leveraging these insights to ensure a seamless upgrade with zero major defects upon launch.

Freelancer | Full-Stack Engineer (Side Projects)

2023.12 - 2024.07

- Designed modern GUIs and implemented low-level hardware control logic using Python; leveraged the UART serial communication protocol to achieve low-latency, real-time command transmission and bidirectional data feedback.
- Independently developed an "Online English Certification Website" utilizing a decoupled MVC architecture with Vue, Laravel, and MySQL; integrated an automated grading mechanism and a dynamic course reservation system to streamline and automate teaching workflows.
- Built a customized official website for a travel agency, leveraging Vue and the Google Drive API ecosystem to replace legacy systems; upgraded data and payment security protocols, slashing the client's maintenance and operational costs by over 80%.

China Broadcasting Corporation (BCC) | Full-Stack Engineer

2021.10 - 2023.11

- Maintained and optimized large-scale .NET Framework & MSSQL systems; integrated third-party payment gateways, automated electronic invoicing, and data encryption verification. Implemented automated background tasks via .NET and enforced cybersecurity controls using VPNs and firewalls.
- Participated in the preparation of the new e-commerce website, leading system analysis (SA) and requirements gathering; defined core business logic, technical specifications, and database migration strategies to ensure a seamless system transition and architectural upgrade.

Educational

National Taiwan University > M.S. in CSIE | GPA: 3.87

- Journal IEEE - Improving the Accuracy Rate of PM2.5 Pollution Forecast Using Q-learning training Time-series Model (IEEE INSPEC Accession Number: 19313037)

National Taiwan University of Science and Technology > B.S. in CSIE | GPA: 3.97 (Ranking: 2/52)

- Intelligent LED-Guided Directional Moving Floor Escape System (C, ARM)
- First Place – National LED Creative Design Competition
- Excellence Award – TiC100 Smart City & IoT Business Model Competition
- First Place – CTCI Foundation Science and Technology Scholarship

(Hardware Control & IoT Security)

- Real-time Payment & Passenger Flow Monitoring System (SA / Architecture Design): Led system and AI architecture planning to shift defenses forward via "pre-payment real-time blocking." Leveraged DDPG reinforcement learning and data fusion to overcome edge log scarcity, incorporating antenna spatial features to trigger instant transaction circuit-breaking. Implemented dual-track forensics using HMM and LSTM-CNN to analyze anomalous sequences and deep packet timing for attack trajectory reconstruction; integrated Policy Gradient and Q-learning for per-log online dynamic learning, enabling self-evolving system capabilities.
- Taipei Medical University - Mobile Medical App & Hardware Integration (COPD Analysis Project - Sole Developer): Developed a mobile medical application using Flutter; deeply integrated C/Arduino firmware via BLE (Bluetooth Low Energy) to achieve low-latency, bidirectional, real-time transmission of medical parameters.
- Commercial Automated Control System: Independently engineered the full-stack system using UWP (.NET Framework); spearheaded UI design and data logic while implementing low-level hardware bidirectional command control through APIs.

(Full-Stack & Systems)

- **【Wei-Mei-You】**Nutrition Management CRM (Sole Developer): Spearheaded UI design, frontend-backend integration, and database schema using .NET Core, Vue.js, and MSSQL; deeply integrated core nutrition management with CRM business logic.
- Real-time Ordering Platform: Developed the platform using Laravel; crafted high-interactivity frontend interfaces and led the architectural design of backend order storage and the "Real-time Settlement & Automated Accounting System."
- **【Traffic or Not】**(Sole Developer): Spearheaded full-stack development using UWP (.NET Framework), driving overall UI design and complex dynamic workflow planning (*Honored with the Global Chinese Sustainable Journalism Award*).
- Virtualization Environment Deployment (Sole Developer): Built and maintained VMware vSphere and Horizon VDI environments, integrating finger-vein biometric authentication to enhance access control; successfully deployed a demo/validation version for the National Chung-Shan Institute of Science and Technology (NCSIST).

(Backend & Data Engineering)

- Yunlin County Government "New Coin" Project: Developed core backend APIs using .NET Core; responsible for rigorous data validation, business logic implementation, and high-performance data restructuring and packaging.
- Antenna Passenger Flow Data Processing: Developed backend APIs using Django (Python) to achieve big data-scale passenger flow data cleaning, multi-dimensional filtering, and anomalous data interception.
- Android Electronic Invoicing App (Sole Developer): Developed a native Android app using Java; dynamically packaged product information and calculated double-barcode formats (1D/QR Code) per Ministry of Finance regulations, achieving deep integration with the government platform.

Trend Micro Taiwan | Network Engineer Assistant**2015.09 - 2019.01**

- Developed an automated inspection tool using Python for**【HomeNetworkSecurity】**; resolved network incompatibility pain points and filtered anomalous logs to enhance home cybersecurity defense.
- Conducted IoT vulnerability and threat research; deeply analyzed security flaws in IoT devices and formulated mitigation strategies to effectively identify malicious domains used in social engineering scams.
- Optimized high-concurrency wireless networks; participated in designing and tuning low-latency wireless services, successfully supporting 150,000 concurrent devices and ensuring high availability in high-density environments.

Journal

- Journal IEEE - Improving the Accuracy Rate of PM2.5 Pollution Forecast Using Q-learning training Time-series Model (IEEE INSPEC Accession Number: 19313037)

Air pollution is a serious problem affecting human life and health. Although predecessors have improved a variety of predictive models of air pollution, the ability to accurately predict air pollution indices is still limited. Time series prediction plays an important role in many fields. Predecessors have tried more artificial neural networks, using linear autoregressive integrated moving average (ARIMA) models with nonlinear neural network (NN) models, and assuming that the time series data is predicted by the air pollution time series in the case where the data is long and no white noise. However, for real-time in short and white noisy time series data, these methods do not guarantee that the prediction error of the NN model is minimized. Therefore, this study proposes a new improvement to the machine learning method. Based on a Reinforcement Learning (RL) method to predict the future PM2.5 value, it uses the Q-learning algorithm in reinforcement learning, based on its state characteristics on the NN model. Select the input between the input dimension and the time delay, calculate the best or near optimal strategy, and use the neural network model to evaluate the computational complexity and accuracy to minimize the prediction error.